Cloud-Based Services

1 Introduction

1.1 Traditional Model

Enterprises traditionally purchased and maintained their own servers, facing:

- High maintenance costs (infrastructure, cooling, power)
- Hardware failures (CPUs, disks, components)
- Difficulty scaling for demand spikes
- Idle infrastructure during low demand periods

1.2 Cloud Computing Model

Applications run on infrastructure managed by a service provider at large data centers. The provider offers:

- Hardware infrastructure
- Platform services (databases, application servers)
- Application software

Major Cloud Providers

Amazon, Microsoft, IBM, Google, and numerous smaller vendors. Amazon pioneered cloud services by offering its internal infrastructure as a service to external users.

2 Cloud Service Models

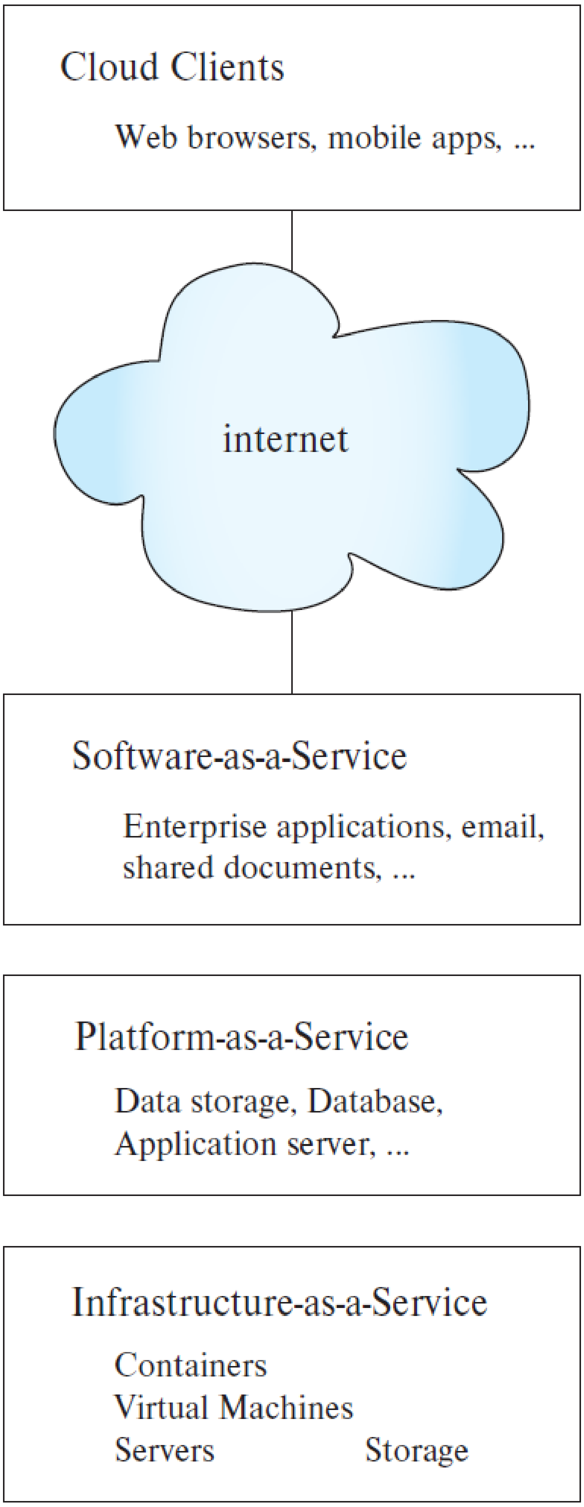

Figure 1: Cloud service models: IaaS, PaaS, and SaaS.

2.1 Infrastructure-as-a-Service (IaaS)

Enterprises rent computing facilities such as physical or virtual machines with disk storage.

Virtual Machines (VMs)

Software-simulated computers that appear as real machines to users. Multiple VMs can run on a single physical server, allowing cloud providers to exploit economies of scale.

Key Benefits:

- **Elasticity:** Expand or contract capacity on demand.
- **Cost Efficiency:** Pay only for resources used.
- **Rapid Deployment:** No need to purchase and install hardware.

Client Responsibilities:

- Install and maintain database systems
- Handle backup and restore operations

2.2 Platform-as-a-Service (PaaS)

Service provider deploys and manages platforms (databases, application servers) that application software uses.

Cloud-Based Data Storage:

- Supports files (MB to GB) or data items (bytes to MB)
- Automatic provisioning of storage and computing power
- Pay-per-use model based on storage and data transfer

Database-as-a-Service (DBaaS):

- Provides SQL or other query language support
- Early offerings: single-node databases
- Modern offerings: parallel database systems

2.3 Software-as-a-Service (SaaS)

Service provider offers complete application software. Clients use web or mobile interfaces without dealing with installation or upgrades.

3 Virtualization Technologies

3.1 Virtual Machines

Developed in the 1960s to share expensive mainframes. Modern benefits:

- Share power and maintenance costs across users
- Easy migration between physical servers
- Quick recovery from hardware failures

Limitation: High overhead—each VM runs a complete operating system.

3.2 Containers

Solve VM overhead problems by sharing the OS kernel while providing isolation.

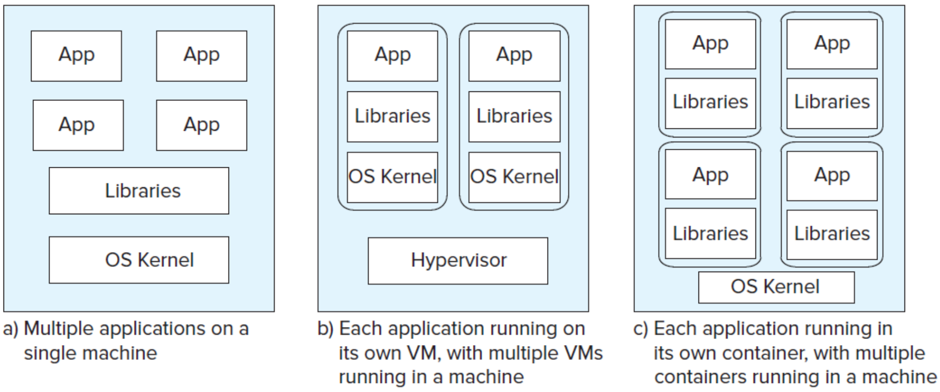

Figure 2: Application deployment alternatives: (a) Multiple apps on one machine, (b) VMs, (c) Containers.

Container Benefits:

- Each container has its own IP address and libraries
- Much lower overhead than VMs
- More containers can run on a single machine
- Quick deployment for elasticity

Isolation:

- Processes within a container can interact via filesystem and IPC
- Inter-container communication only via network

4 Microservices Architecture

Applications built as collections of small services, each offering a network API.

Microservices

An architecture where applications are composed of multiple independent services, each running as a separate process. Containers provide ideal low-overhead execution for microservices.

4.1 Docker & Kubernetes

Docker: Widely used container platform.

Kubernetes: Advanced orchestration platform providing:

- Declarative container specification
- Automatic deployment and linking of containers
- **Pods:** Multiple containers sharing storage and network
- **Elasticity Management:** Automatic scaling
- **Load Balancing:** Distributes requests across container replicas

5 Benefits and Limitations

5.1 Benefits

- Eliminates need for large system-support staff
- No large upfront capital investment
- On-demand resource scaling
- Economies of scale from large vendor clusters

5.2 Limitations

Security & Legal Risks

Data Control: Client data held by another organization.

- Security breaches may expose client data
- No direct control over vendor security
- Geographic data storage affects legal jurisdiction
- Privacy laws vary by country (e.g., EU vs. US)
- Vendors may be required to release data to governments

Market Reality

Despite drawbacks, the benefits of cloud services are compelling enough to drive rapid market growth. Many enterprises accept the risks for the substantial cost and operational advantages.